

Exercise 4, part ii

```
a = 10;b = 4;c = 2;  
lowlim = {-a,@ylo,-2};  
uplim = {a,@yup,-1.1};  
gsimp(@conc,lowlim,uplim,10)
```

```
ans = 6.1171e+03
```

The calculated mass of the dense non-aqueous phase liquid (DNAPL) is approximately 6117.1 mg.

```
function [v] = ylow(X)  
x=X(1);  
a = 10;b = 4;  
v = -b*sqrt(1-((x/a)^2));  
end  
  
function [v] = yup(X)  
x=X(1);  
a = 10;b = 4;  
v = b*sqrt(1-((x/a)^2));  
end  
  
function [c] = conc(X)  
z=X(3);  
C0=100;  
c = C0/sqrt(1+z^2);  
end  
function [I] = gsimp(f,a,b,n)  
m=length(a); x=zeros(1,m); k=1;  
% convert limits to cell array if numbers are given  
if (isnumeric(a))  
    a=num2cell(a);  
end  
if (isnumeric(b))  
    b=num2cell(b);  
end  
% call recursive integration function  
I=ksimp(f,x,a,b,n,k);  
end  
  
% recursive simpson function  
function [I] = ksimp(f,x,a,b,n,k)  
I=0; m=length(a);  
if isnumeric(a{k})  
    ak=a{k};  
else  
    ak=feval(a{k},x);  
end  
if isnumeric(b{k})  
    bk=b{k};  
else  
    bk=feval(b{k},x);
```

```

end
h =(bk-ak)/n;
for i=0:n
    x(k)=ak+i*h;
    if (k == m)
        if (i == 0 || i == n)
            I=I+feval(f,x);
        elseif (mod(i,2) == 1)
            I=I+4*feval(f,x);
        elseif (mod(i,2) == 0)
            I=I+2*feval(f,x);
        end
    else
        if (i == 0 || i == n)
            I=I+ksimp(f,x,a,b,n,k+1);
        elseif (mod(i,2) == 1)
            I=I+4*ksimp(f,x,a,b,n,k+1);
        elseif (mod(i,2) == 0)
            I=I+2*ksimp(f,x,a,b,n,k+1);
        end
    end
end
I=I*h/3;
end

```